

class10

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The PDB database

The **Protein Data Bank** (<http://www.rcsd.org/>) is the main repository of biomolecular structure data. Let's see what is in it:

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

X-Ray is 81% and Electron Microscopy is 13%. >Q2: What proportion of structures in the PDB are protein?

```
stats <- read.csv("pdb_stats.csv", row.names = 1)
protein_rows <- grep("^Protein", rownames(stats))
protein_count <- sum(stats[protein_rows, ])
total_count <- sum(stats)
protein_proportion <- protein_count / total_count
protein_proportion
```

```
[1] 0.979118
```

```
stats <- read.csv("pdb_stats.csv", row.names=1)
stats
```

	X.ray	EM	NMR	Integrative	Multiple.methods	Neutron
Protein (only)	178795	21825	12773	343	226	84
Protein/Oligosaccharide	10363	3564	34	8	11	1
Protein/NA	9106	6335	287	24	7	0
Nucleic acid (only)	3132	221	1566	3	15	3
Other	175	25	33	4	0	0
Oligosaccharide (only)	11	0	6	0	1	0
	Other	Total				
Protein (only)	32	214078				
Protein/Oligosaccharide	0	13981				
Protein/NA	0	15759				
Nucleic acid (only)	1	4941				
Other	0	237				
Oligosaccharide (only)	4	22				

```
n.sums <- colSums(stats)
n <- n.sums/n.sums["Total"]
round(n, digits=2)
```

	X.ray	EM	NMR	Integrative
	0.81	0.13	0.06	0.00
Multiple.methods	Neutron		Other	Total
	0.00	0.00	0.00	1.00

What is the total number of entries in the PDB

```
n.sums["Total"]
```

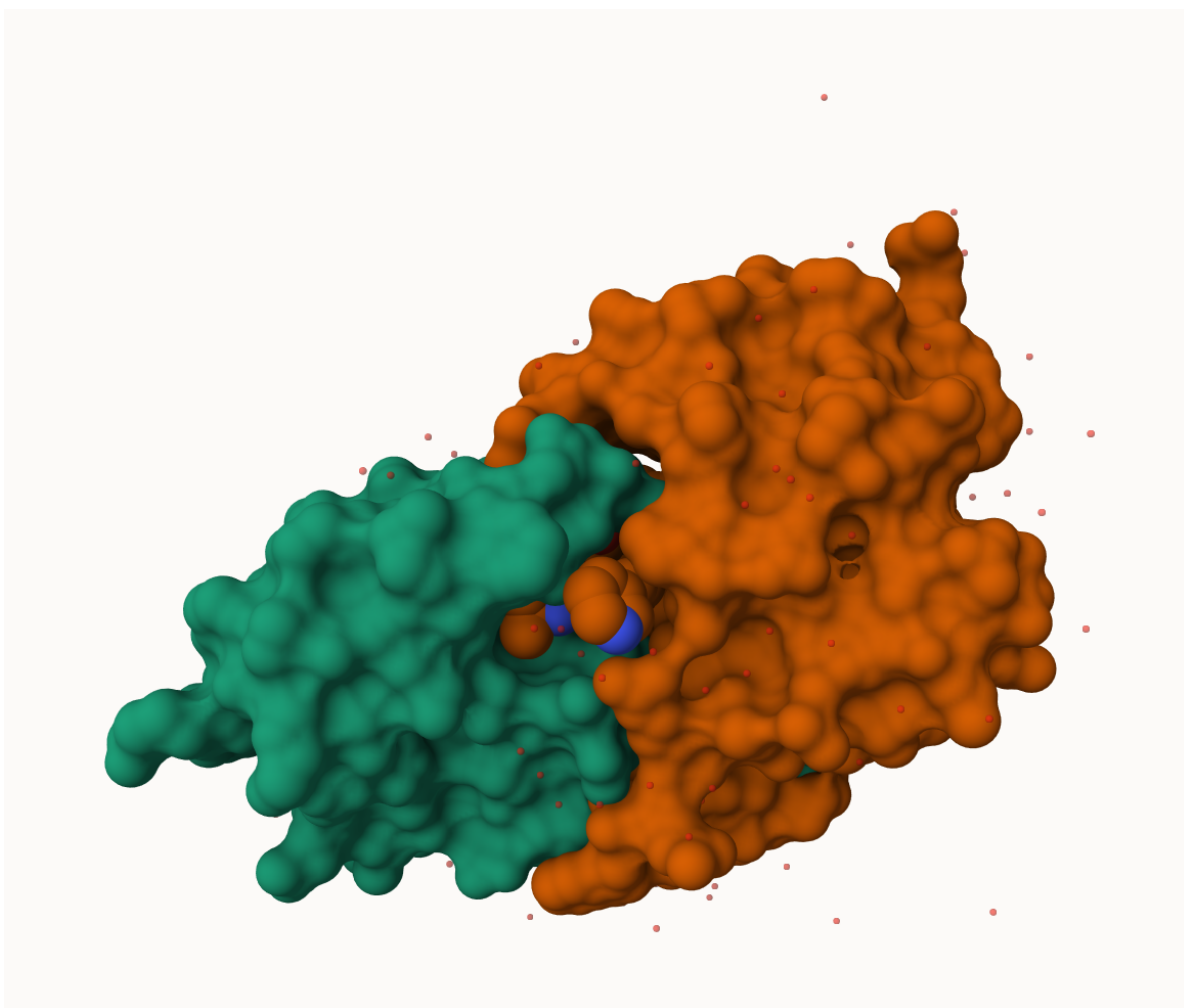
```
Total
249018
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

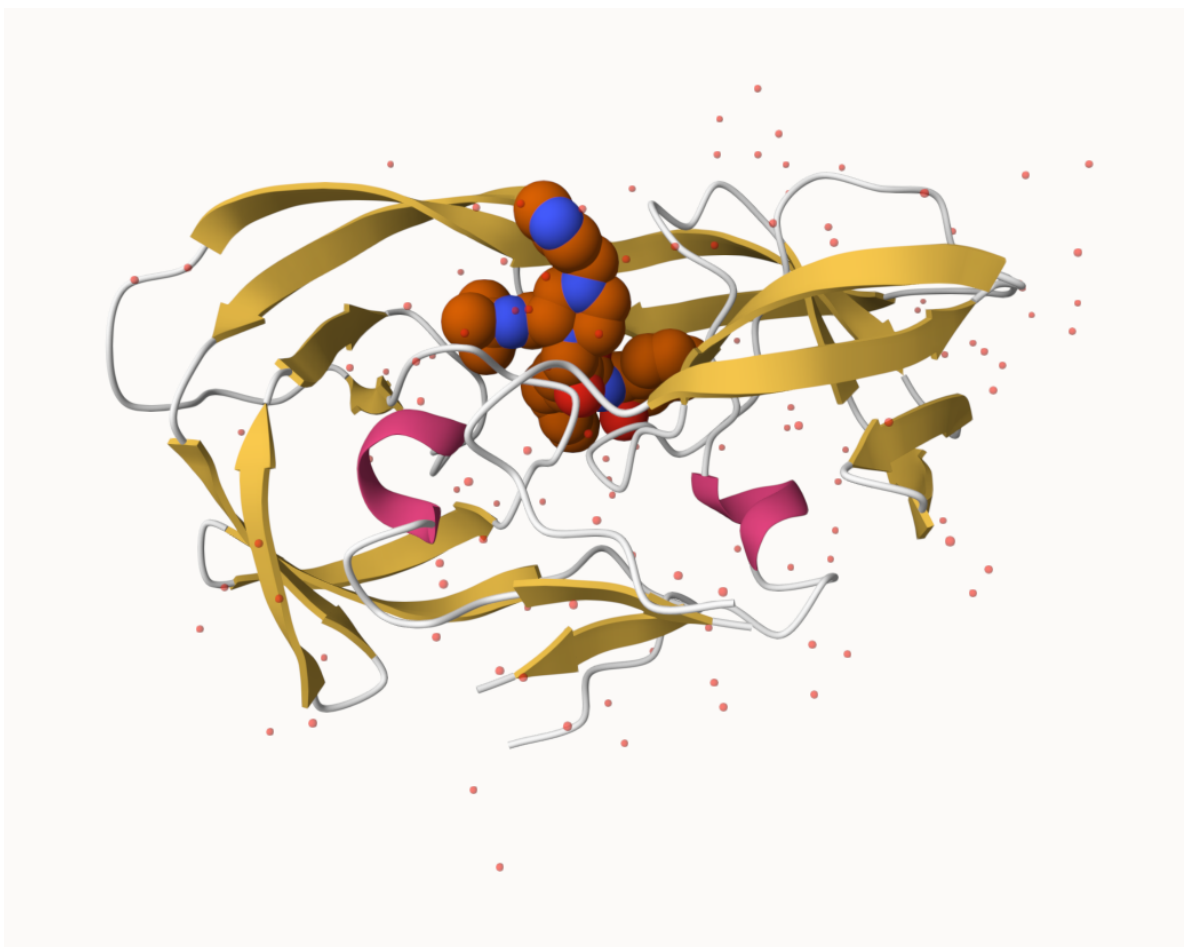
4,940 structures

Using Molstar

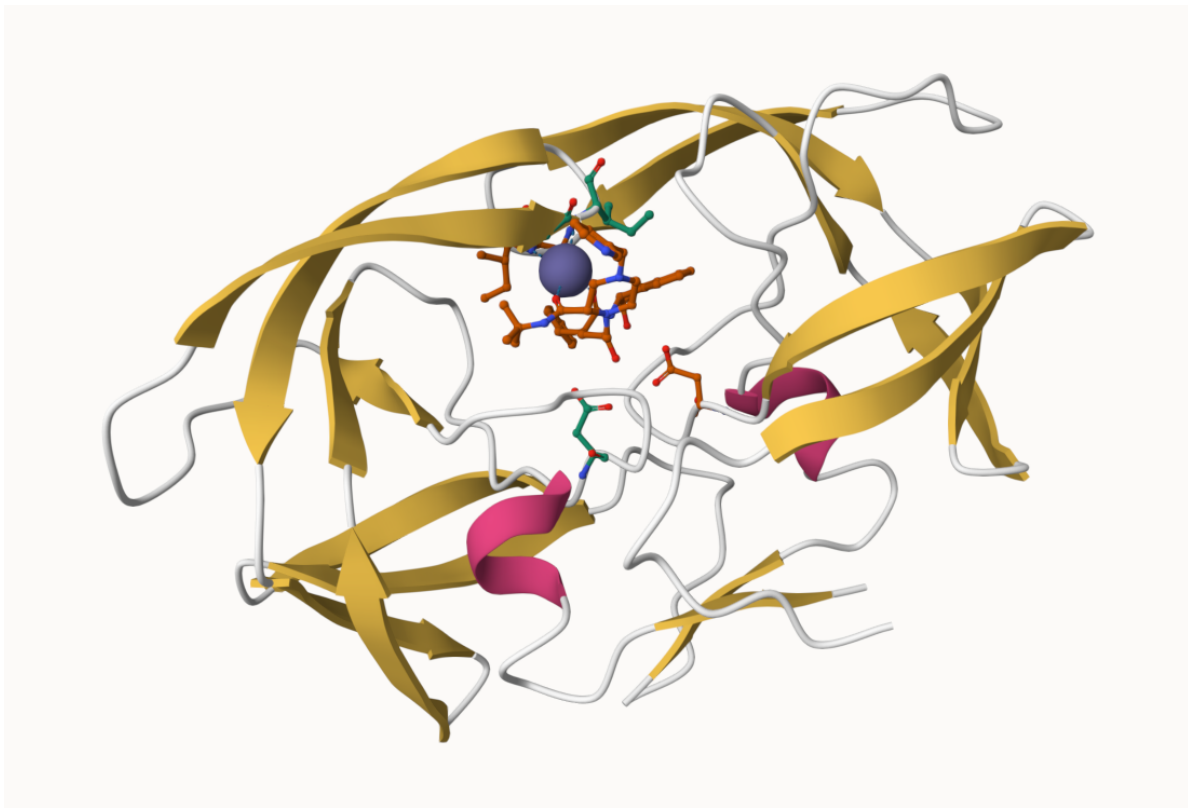
We can use the main [Molstar viewer online](#):



>Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary structure, showing the inhibitor (ligand) in spacefill.



Q. One final image showing catalytic APS 25 and the all important active site water molecule as spacefill.



The Bio3D package for structural bioinformatics

```
library(bio3d)  
hiv <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
hiv
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
 Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
 Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
 QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
 ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
 VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
 calpha, remark, call

`head(hiv$atom)`

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

`pdbseq(hiv)`

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
"P"	"Q"	"I"	"T"	"L"	"W"	"Q"	"R"	"P"	"L"	"V"	"T"	"I"	"K"	"I"	"G"	"G"	"Q"	"L"	"K"
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
"E"	"A"	"L"	"L"	"D"	"T"	"G"	"A"	"D"	"D"	"T"	"V"	"L"	"E"	"E"	"M"	"S"	"L"	"P"	"G"
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
"R"	"W"	"K"	"P"	"K"	"M"	"I"	"G"	"G"	"I"	"G"	"G"	"F"	"I"	"K"	"V"	"R"	"Q"	"Y"	"D"
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80

```

"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 1
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"

```

Let's try out the new **bio3dview** package that is not yet a CRAN.

```

library(bio3dview)

sele <- atom.select(hiv, resno=25)
#view.pdb(hiv, backgroundColor = "pink",
#         highlight = sele,
#         highlight.style = "spacefill")

```

Prediction of Protein flexibility

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

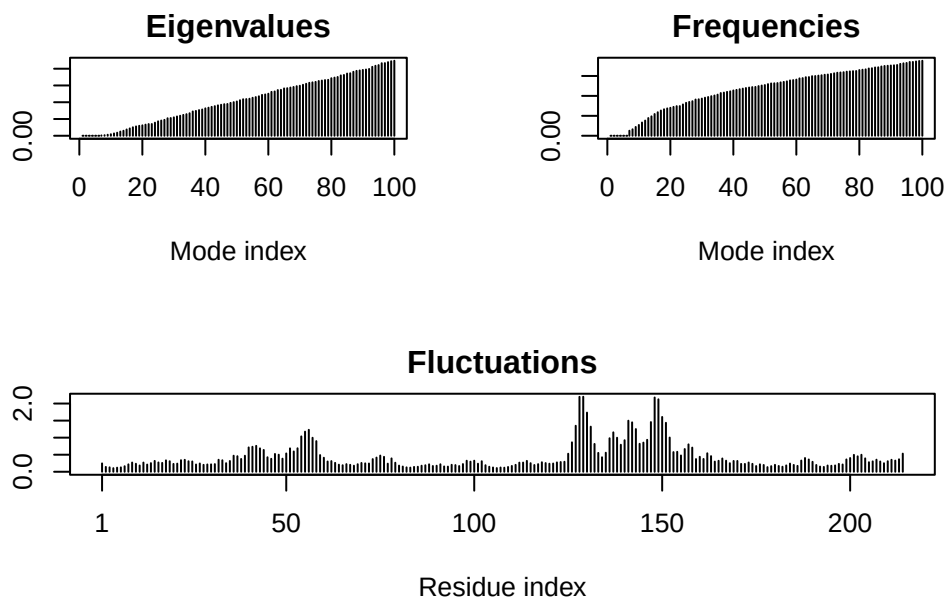
```
m <- nma(adk)
```

```

Building Hessian...      Done in 0.019 seconds.
Diagonalizing Hessian... Done in 0.079 seconds.

```

```
plot(m)
```



Write out our results as a wee trajectory movie:

```
#mktrj(m, file="results.pdb")
```

```
#view.nma(m)
```

Comparative protein structure analysis with PCA

We start with a database id "1ake_A"

```
library(bio3d)
id<-"1ake_A"
aa<- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.


```
aa
```

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

```
blast<- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = V62T1PD1016

.....

Reporting 96 hits

Have a wee peak:

```
head(blast$hit.tbl)
```

```
queryid subjectids identity alignmentlength mismatches gapopens q.start
```

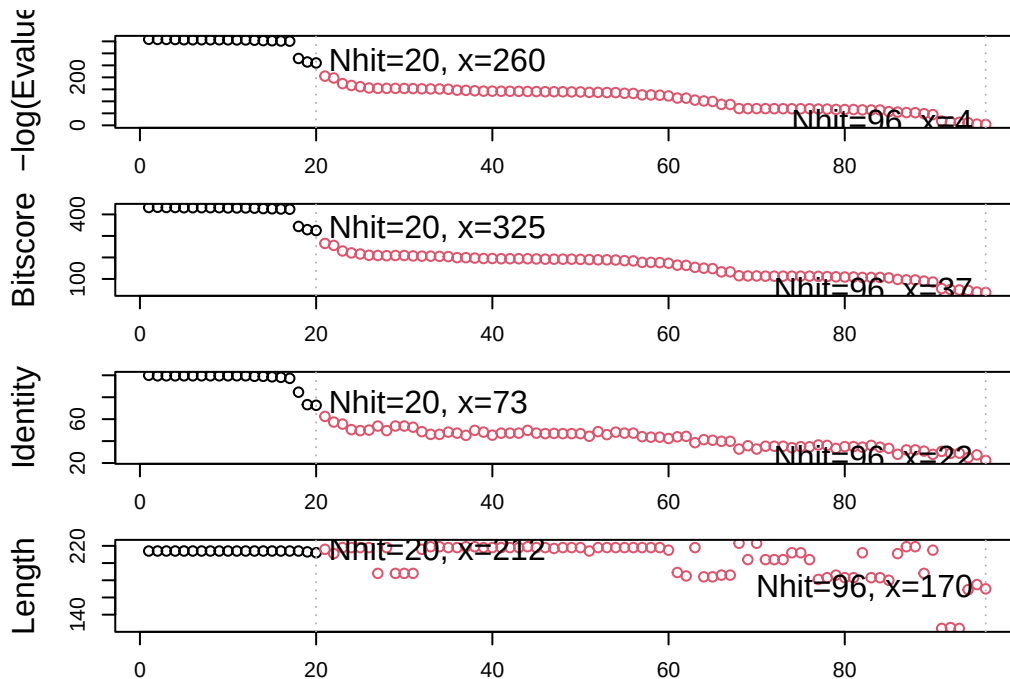
1	Query_6448369	1AKE_A	100.000			214	0	0	1
2	Query_6448369	8BQF_A	99.533			214	1	0	1
3	Query_6448369	4X8M_A	99.533			214	1	0	1
4	Query_6448369	6S36_A	99.533			214	1	0	1
5	Query_6448369	9R6U_A	99.533			214	1	0	1
6	Query_6448369	9R71_A	99.533			214	1	0	1

	q.end	s.start	s.end	evalue	bitscore	positives	mlog.evalue	pdb.id	acc
1	214	1	214	1.79e-156	432	100.00	358.6211	1AKE_A	1AKE_A
2	214	21	234	2.93e-156	433	100.00	358.1283	8BQF_A	8BQF_A
3	214	1	214	3.20e-156	432	100.00	358.0401	4X8M_A	4X8M_A
4	214	1	214	4.71e-156	432	100.00	357.6536	6S36_A	6S36_A
5	214	1	214	1.05e-155	431	99.53	356.8519	9R6U_A	9R6U_A
6	214	1	214	1.24e-155	431	99.53	356.6856	9R71_A	9R71_A

```
hits <- plot(blast)
```

```
* Possible cutoff values:    260 3
    Yielding Nhits:         20 96
```

```
* Chosen cutoff value of:    260
    Yielding Nhits:         20
```



Peak at our “top hits”

```
head (hits$ pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

Now we can download these “top hits” these will all be ADK structures in the PDB database.

```
files<- get.pdb(hits$ pdb.id, path="pbds", split = TRUE, gzip = TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/9R6U.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/9R71.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/8Q2B.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/8RJ9.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pbds", split = TRUE, gzip = TRUE):  
pbds/4X8H.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/6HAP.pdb.gz exists. Skipping download

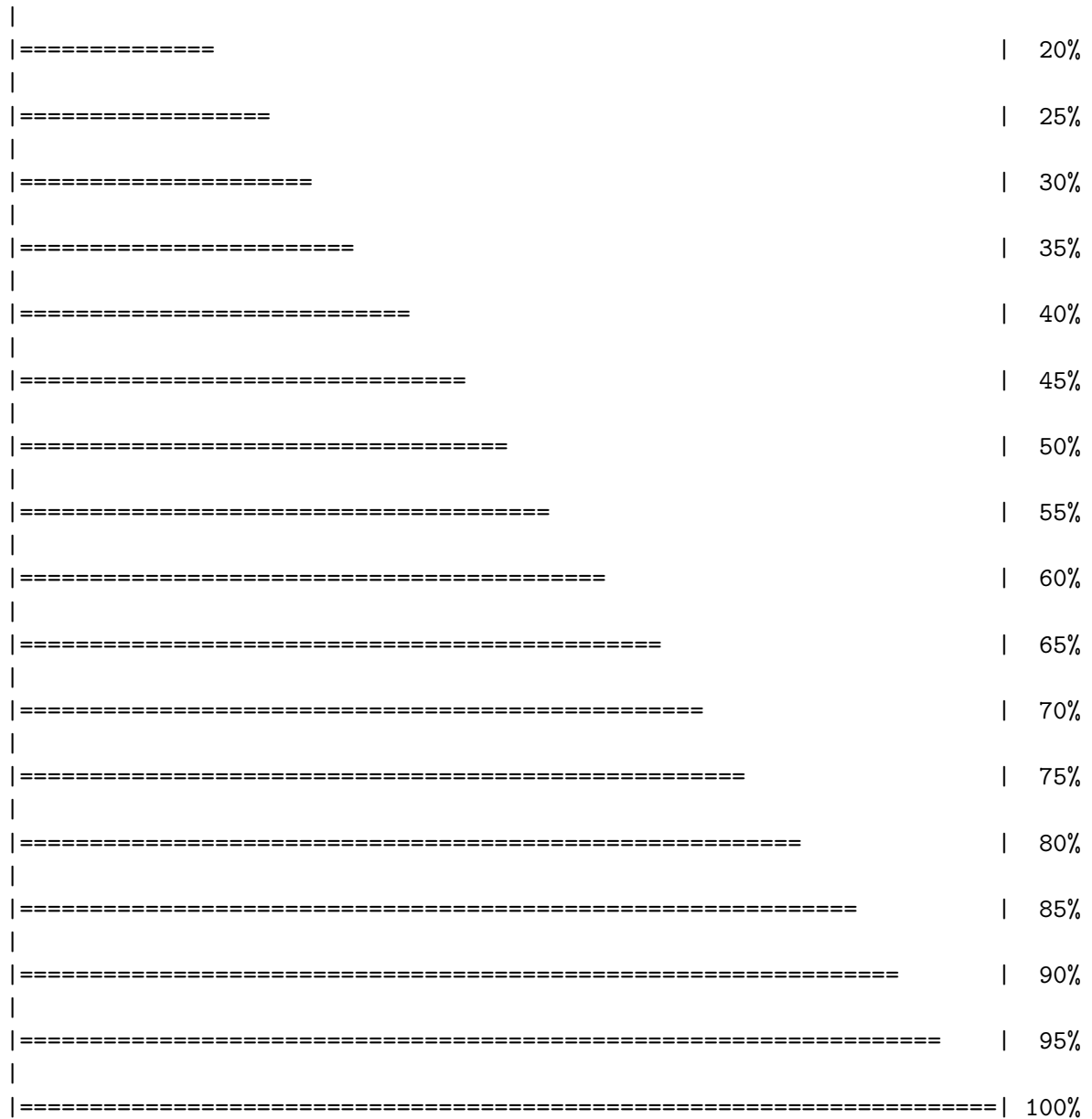
Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pbds", split = TRUE, gzip = TRUE):
pbds/4NP6.pdb.gz exists. Skipping download

	0%
====	5%
=====	10%
=====	15%



We need one package from BioConductor. To set this up we need to first install a package called “**BiocManager**” from CRAN.

Now we can use the `install()` function from the package like this: `BiocManager::install("msa")`

```
pdb<- pdbaln(files, fit = TRUE, exe="msa")
```

Reading PDB files:

```

pbds/split_chain/1AKE_A.pdb
pbds/split_chain/8BQF_A.pdb
pbds/split_chain/4X8M_A.pdb
pbds/split_chain/6S36_A.pdb
pbds/split_chain/9R6U_A.pdb
pbds/split_chain/9R71_A.pdb
pbds/split_chain/8Q2B_A.pdb
pbds/split_chain/8RJ9_A.pdb
pbds/split_chain/6RZE_A.pdb
pbds/split_chain/4X8H_A.pdb
pbds/split_chain/3HPR_A.pdb
pbds/split_chain/1E4V_A.pdb
pbds/split_chain/5EJE_A.pdb
pbds/split_chain/1E4Y_A.pdb
pbds/split_chain/3X2S_A.pdb
pbds/split_chain/6HAP_A.pdb
pbds/split_chain/6HAM_A.pdb
pbds/split_chain/8PVW_A.pdb
pbds/split_chain/4K46_A.pdb
pbds/split_chain/4NP6_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pbds/split_chain/1AKE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pbds/split_chain/8BQF_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pbds/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pbds/split_chain/6S36_A.pdb

```

```

PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pbds/split_chain/9R6U_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pbds/split_chain/9R71_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pbds/split_chain/8Q2B_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pbds/split_chain/8RJ9_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pbds/split_chain/6RZE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pbds/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pbds/split_chain/3HPR_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pbds/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pbds/split_chain/5EJE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pbds/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pbds/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pbds/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pbds/split_chain/6HAM_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pbds/split_chain/8PVW_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pbds/split_chain/4K46_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20  name: pbds/split_chain/4NP6_A.pdb

```

Let's have a wee peak at our structures after "fitting" or superimposing:

```

library(bio3dview)
#view.pdb(pdb)

```

```

#view.pdb(pdb, colorScheme = "residue")

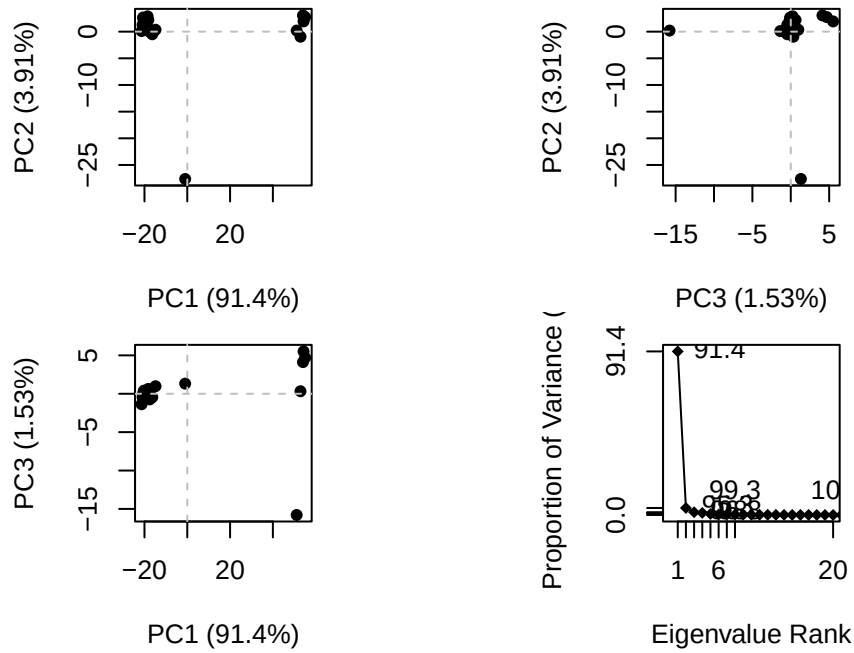
```

We can run functions like `rmsd()`, `rmsf()` and the best `pca()`

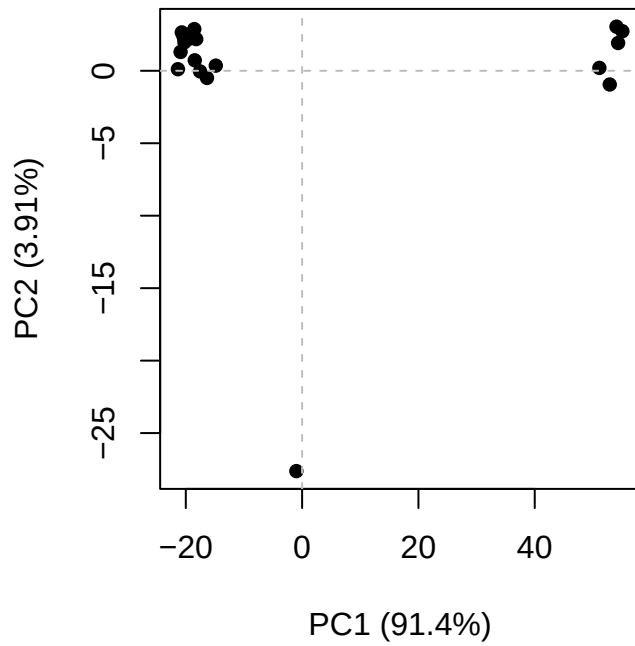
```

pc.xray <- pca(pdb)
plot(pc.xray)

```



```
plot(pc.xray, 1:2)
```



Finally, let's make a wee movie of the major "motion" or structural differences in the dataset


```
#mktrj(pc.xray, file="results.pdb")
```